

Fig. 3. GeoTex Bake10 lowest-absolute-error example at held-out camera 004. The Stone object has the lowest existing eleven-view FG-LPIPS among seven display-cleared assets, but Late-high minus Fixed Mean is only -0.000464 **FG-LPIPS** ($+0.001$ **dB FG-PSNR). GT-256 reference self-bake, Fixed Mean, and Late-high use one common crop and the same mesh/UVs. This outcome-selected example separates absolute error from schedule benefit; it does not show a general effect or faithful material reproduction. The complete ten-object effect distribution is in Supplementary Section S5.**

Table 5. Unseen-view FG-LPIPS primary contrasts for the two bake studies.

Study (N)	Δ FG-LPIPS	95% CI	p
GeoTex (10)	−0.002652	[−0.004144, −0.001367]	0.00195
GSO (20)	+0.000084	[−0.000103, +0.000286]	0.788

Late-high minus Fixed Mean. GeoTex: exact paired sign-flip. GSO: original-family Holm16-adjusted paired sign-flip. Separate input and calibration protocols are not pooled; non-detection is not equivalence.

image/bake LPIPS-rank correlation is 0.601 (object interval $[0.440, 0.733]$). A post hoc sensitivity restricted to the seven original scale/time controls gives 0.396 ($[0.175, 0.582]$), with the minimum-mean LPIPS condition changing on 11/20 objects. These are positively associated rankings with changing point-estimate winners, not general rank reversal or statistically resolved winner changes. The GT self-bake has mean LPIPS 0.0602 and PSNR 19.36 dB; average complete-bilinear-footprint coverage is about 78.7%. All foreground remains scored and GT errors are not subtracted. These losses contextualize the small method effects without defining a theoretical quality bound.

Incoming-view consistency. The bake also measures cross-camera color variance at UV texels receiving multiple source views. A constant-black control has zero variance despite its incorrect appearance, showing that this diagnostic measures incoming-view disagreement rather than fidelity. Late-high has a lower mean disagreement than Fixed Mean (-0.002448), while Fixed High has still lower disagreement and higher baked LPIPS. Resampling and overlapping UVs also affect this measure; it does not validate seams or PBR. Full controls and contrasts are in Supplementary Section S5.

Native controls and protocol sensitivity. Subsequently added Native residual and No Adapter outputs on the same ten GeoTex objects are exploratory. They retain matched geometry and GT coverage, and the repeated Constant 1.0 (capped) outputs are bitwise identical. Their apparent PSNR/LPIPS gains for the schedules do not separate under the fifteen-test exploratory family. The complete data are in Supplementary Section S5.

A supporting fifty-object comparison uses a newly trained frozen GeoTex checkpoint with 512-pixel supervision, actual positive/negative reference conditions and CFG 2. Late-high minus Fixed Mean has nonreference PSNR -1.184 dB (interval $[-1.587, -0.849]$) and LPIPS $+0.00266$ ($[-0.00003, +0.00640]$; adjusted $p = 0.080$). C3 has no corrected advantage over Fixed Mean or Fixed High. This previously studied cohort and changed checkpoint/protocol supply a sensitivity boundary, not an independent backbone confirmation. Training details and full-v29 comparisons are in Supplementary Section S8.

Native-pipeline quality boundary. The recorded full-v29 ControlNet pipeline is a stronger deployment comparator than geometry-disabled output from the new HF-based checkpoint. On the same fifty inputs and five nonreference cameras, the new geometry checkpoint trails full v29 by 2.16 dB FG-PSNR (paired interval $[-3.15, -1.27]$) and has FG-LPIPS higher by 0.0341 ($[0.0235, 0.0464]$); these belong to the separately frozen four-endpoint Holm family. This is a matched-input deployment comparison, not a scale-only comparison: base weights and geometry implementations differ, and the common 50-step, seed-42, CFG-2 setting is not the official command-line default. In an exploratory follow-up that matches the exported v29 base UNet, unscaled geometry improves LPIPS over disabled geometry by 0.0131 yet remains 0.0159 worse than full v29 (paired interval $[0.0085, 0.0237]$). Geometry architecture and training history still differ, and the same previously inspected objects were reused. Thus a local control benefit can coexist with a larger complete-system quality deficit; neither contrast establishes TCAS deployment superiority. Full conditions, tests and grids are in Supplementary Section S8.

4.5. Sensitivity Across Native Conditioning Interfaces

The auxiliary studies ask related scheduling questions under different native conditioning operators and camera contracts.



Fig. 3. GeoTex Bake10 appearance at held-out camera 004. Each row shows a GT-256 reference self-bake, Fixed Mean, and Late-high under a shared mesh/UV and within-object crop. Tire is selected by median source-foreground Laplacian variance among seven display-cleared assets. The carved font is an explicitly post-outcome example with the largest existing eleven-view Late-high minus Fixed Mean FG-LPIPS reduction among those seven (−0.00781); its FG-PSNR also decreases by 0.271 dB. The selected examples illustrate appearance mismatch and an endpoint tradeoff, not cohort prevalence or a general schedule benefit. Source images are 512×512 and displayed without method-specific color edits. The full ten-object effect distribution and scores are in Supplementary Section S5.

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is not a GT self-bake. Object, camera, and display limits are detailed in Supplementary Sections S5 and S4.

Across all twelve GSO conditions, mean within-object image/bake LPIPS-rank correlation is 0.601 (object interval $[0.440, 0.733]$). A post hoc sensitivity restricted to the seven original scale/time controls gives 0.396 ($[0.175, 0.582]$), with the minimum-mean LPIPS condition changing on 11/20 objects. These are positively associated rankings with changing point-estimate winners, not general rank reversal or statistically resolved winner changes. The GT self-bake has mean LPIPS 0.0602 and PSNR 19.36 dB; average complete-bilinear-footprint coverage is about 78.7%. All foreground remains scored and GT errors are not subtracted. These losses contextualize the small method effects without defining a theoretical quality bound.

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